AI-Assisted Personalized Learning and Student GPA Performance in Higher Education

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ABSTRACT:

This longitudinal study evaluates how structured AI tutoring and self-regulated AI usage frequency affect semester Grade Point Average (GPA) across disciplines. Sample size: N = 510 students across Computer Science, Economics, and Social Sciences.

KEY EMPIRICAL FINDINGS:

1. GPA Improvement: Moderate, structured AI usage (4 to 8 hours/week) was associated with an average GPA increase of +0.34 points (on a 4.0 scale) compared to non-users.
2. Optimal Usage Pattern: Students using AI 'Frequently' (3-5 days/week) achieved the highest mean GPA (3.42), outperforming both 'Rarely' (2.88) and excessive passive users.
3. Positive Correlation: Overall guided AI study hours showed a moderate positive correlation with academic GPA ($r = 0.58$, $p < 0.001$).